

Structure and function of wildfire and mountain pine beetle forest boundaries

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Landscape boundaries, in particular those created by natural disturbances, are fundamental structures in landscape functioning. At the stand scale, forest boundaries show a wide range of characteristics varying from sharp to diffuse, wavy to straight, enclosed areas (patches) to open ones; all of these have different effects of forest vegetation. The objectives of this study were 1) to characterize the structure of forest boundaries (width, steepness, heterogeneity) and 2) to investigate forest vegetation functional response in the boundary zones created by wildfires and mountain pine beetle (MPB) outbreaks. We studied 11 mountain pine beetle and 7 wildfire boundary zones (4–7 yr since disturbance) using field-based two-dimensional grids in southern central British Columbia (Canada). Boundary zone delineation was determined through spatially constrained clustering algorithms at three sampling unit resolutions (5×5 m, 10×10 m, and 20×20 m). Then, to characterize boundary width and shape, we developed two new boundary indexes (maximized and minimized boundary zones). The identified boundary widths ranged from 0 to 127 m with a mean width of 51 m at the 20×20 m resolution for both fire and MPB. Although the widths were comparable between disturbance types, fires generally had steeper boundaries (more pronounced) than MPB, largely due to higher peak tree mortality within the disturbances (89 vs 43%). Most of the forest vegetation response variables (understory vegetation diversity, plot-level species richness, evenness, and multivariate community measures) in the boundary zones tended not to be intermediate in value between the intact forest and disturbance area, in spite of intermediate tree mortality. Tree mortality heterogeneity in the boundary zones was often highest in fires and was equal to the internal heterogeneity in MPB disturbances. Using historical natural disturbance patterns as a conservation strategy, this study proposes that forest management should create boundary complexity (width and shape), thereby creating landscape heterogeneity similar to landscapes influenced primarily by natural disturbances.

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Over the last decades, forests have changed from continuous landscapes to fragmented woodlots. A by-product of fragmented landscape is the predominant amount of edges, boundaries, as structural landscape features. These vegetation boundaries have high degrees of structural heterogeneity resulting in important fundamental

functions in landscapes (Lidicker 1999) such as large rates of ecological change through space (Wiens et al. 1985, Hansen and di Castri 1992). Moreover, their ecological influence may be disproportionate to their area on the landscape (Laurance and Yensen 1991, Holland et al. 1991, Chen et al. 1992, Matlack 1993,

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Voller 1998). This is mostly due to the fact that boundary function changes depending on its structure (Cadenasso and Pickett 2000, Cadenasso et al. 2003).

A boundary is a complex spatial structure whose characterization can change dramatically from a sharp edge to a diffuse and large scale ecotone depending on the scale of analysis (Gosz 1993). This rich suite of boundary characteristics is often depicted in a polygon-based land-use map as a series of linear, one-dimensional features dissecting the landscape. From such mapping representations, we cannot discriminate actual one-dimensional linear boundaries from two-dimensional landscape features that have unique ecological characteristics. This is unfortunate as there are functional differences between boundary types such as gradual and sharp boundaries (Fagan et al. 1999). For instance, Ratti and Reese (1988) demonstrated that there was higher nest predation at sharp boundaries compared with feathered edges. Possible explanations for different function include the presence of spatial heterogeneity in diffuse boundaries that benefits many organisms, the close juxtaposition of multiple habitat types, and creation of unique habitat that is not simply the average of the intact forest and disturbance (Forman 1995). In the case of nest predation, spatial heterogeneity allows for nest hiding and many feeding opportunities (Bowman and Harris 1980, St-Louis et al. 2004). Boundaries that occur at the periphery of natural disturbances may be softer than a linear depiction suggests and may present spatial heterogeneity. To understand forest boundaries, the variety of boundaries created by natural processes must be objectively described and their effects on forest vegetation studied.

Natural disturbances are agents of tree mortality, often creating an edge, and have become a key ecological focus of conservation and forest management. A widespread suggestion for a coarse filter conservation template for stand and landscape forest management is to base management on the spatial and temporal patterns and processes in natural disturbances (Galindo-Leal and Bunnell 1995, Christensen et al. 1996, Bergeron et al. 1999, Swetnam et al. 1999, Bengtsson et al. 2000). Many instances of forest management to date have created landscape homogeneity and are not representative of natural fires (Turner et al. 1994) or other natural disturbances such as insect outbreaks. Many applications of implementing natural disturbance ideas into forest management have begun (Anon. 1995, Cissel et al. 1999). Such edge diffuseness (gradual boundary), however, has rarely been directly manipulated in forest management, except in the context of making edges windfirm (able to persist even with strong winds).

Fires and some types of large insect outbreaks (e.g. mountain pine beetle) share some general features of their spread behavior, like sites of origination and movement outwards non-randomly, but they may differ

in the way the actively moving disturbance agent stops: fires may stop by cooler weather but encountering a lack of host may act to stop fires or insects (McIntire 2004). This slowing or ending of the spread may or may not happen rapidly in time, thereby creating a boundary that may or may not be "abrupt" in space (Forman 1995). Here, a boundary zone is defined as the two dimensional area that contains intermediate mortality between the most severe parts of a disturbance event and the area immediately outside the event.

The objectives of this study were to determine the structure of the boundary zone created by fires and mountain pine beetle (MPB) by delineating and comparing them, and to examine the vegetation response to these boundary zones. Several specific questions are addressed that have ecological and management implications: do MPB and fires differ in their spatial patterns (width, steepness, and heterogeneity) at their edges? Is the boundary zone more heterogeneous with respect to living and dead trees than the intact forest or disturbance core? What are the vegetation and tree foliage responses to this boundary? Throughout all of these questions, we examine the data at three sampling unit resolutions (5×5 m, 10×10 m and 20×20 m) to see how resolution affects the answers. To address these issues, we need to use a boundary detector that can delineate both boundary lines and boundary zones and also assess boundary width variability. Here, we present a methodology for delimiting boundary zones and we develop a new approach to determine boundary width variability.

Methods

Study area

Research was conducted in the summer of 2001 in the Arrow Lakes and Kootenay Forest Districts, in steep (slope angles between 10° and 30°) conifer forests of south central British Columbia. All MPB sites were dominated by lodgepole pine *Pinus contorta* and fire sites were dominated by lodgepole pine, subalpine fir *Abies lasiocarpa*, white $\times$ Engelmann spruce hybrids *Picea glauca* $\times$ *Picea engelmannii*, or Douglas-fir *Pseudotsuga menziesii*. Individual MPB sites had secondary species that could include any of the above species. According to the British Columbia biodiversity guidebook (Anon. 1995), these zones are characterized by stand replacing fire disturbance regimes, ranging from 150 to 400 yr (Natural Disturbance Types 1 and 2, rare or infrequent stand initiating events). All disturbances occurred between 1993 and 1997.

We sampled five fires that were all lightning caused, had extensive areas that were not salvaged, were amenable to being sampled (i.e. no slopes $> 50^\circ$), occurred at elevations between 1400 and 1900 m, and were in the

province's fire database (Anon. 2001a). All fires had had active fire suppression efforts; however, site selection within each fire was constrained to sites where there was no or minimal evidence of local suppression efforts (e.g. Pulaski lines [fire guards], red retardant embedded in downed wood, evidence of trees cut during the fire). Fire boundaries examined in this study were along the side flanks of the fire event that moved up the slope and were located horizontally across the mountain slope. This was done for very practical reasons, as many of the fires had large fire breaks that were successful at stopping the fire at the top of the burns. Thus, there were no quasi-natural edges at the top of the fires.

We sampled eleven MPB sites that had the following characteristics: mature stands (60 yr < tree age at breast height < 150 yr); canopy trees at least 20 cm, DBH; southerly aspect slopes (97°–256°); and, elevation between 1150 and 1550 m. These sites comprised all the sites that were available to be sampled based on the following criteria: sites that were recorded as “severity 3” in the forest health database (Anon. 2001b), that were within the study area and biogeoclimatic zones. MPB stands were overall of similar character due to specific habitat selection by the insect disturbance (mature lodgepole pine).

Sampling transects

All sampling occurred using two-dimensional sampling transects that ranged from 60 to 170 m long and 40 to 60 m wide (Fig. 1; Table 1). Seven two-dimensional sampling transects were established at the edges of five different fires (sites F12 and F20, and F17 and F21 were from the same fires, however, the two transects were separated by at least 1 km from each other). Eleven similar transects were established in the eleven MPB outbreaks. Each transect had a field-estimated 20% of the plots inside the disturbance core and a field estimated 20% outside the disturbance (near zero tree mortality). The overall length and width of each

sampling transect was determined by the variability of the stand being measured. The two transects that were wider than 40 m had low mortality (M09) and very low tree densities (F17). Random placement of transects was done in a three-step process. First, MPB outbreaks or fires, were found using fire and insect databases (Anon. 2001a, b). Several spatially random points were located within each outbreak or fire using a GIS and random point generator (Anon. 1999). Second, one of the random points was visited and assessed for the following: was there actually tree mortality, and was the site free or virtually free of fire suppression. When these conditions were not satisfied, a second (and third, etc.) random point within that disturbance event was visited and the assessment was repeated until a random point that fulfilled these conditions was found. Third, a line perpendicular to the closest mapped boundary was examined on the ground for at least 500 m or until evidence of the disturbance had virtually disappeared (i.e. no fire scarring, and no obvious clusters of MPB killed trees). In all cases but one (F12), continued sampling along the transect outward from the disturbance did not cause reentry into more disturbed areas. For fires, this line was invariably along a contour line across the mountain slope crossing the side flank of the fire that had moved upslope.

Data collection

Each transect was sampled at a 25 m² resolution (5 × 5 m plots). All data analyses were performed on these 5 × 5 m plots and were repeated on aggregates of 4 (10 × 10 m) or 16 (20 × 20 m) of these small plots to address multiple resolutions of spatial pattern. All living and dead trees >10 cm diameter at breast height were measured (diameter at breast height, species, height, alive/dead, cause of mortality). From these, basal area was calculated (the sum of the surface area of tree trunks at breast height) for each plot. The median number of trees per 5 × 5 m plot was 2, with a range from 0 to 12.

Fig. 1. Sampling design. All transects had this form, however, the longitudinal length and latitudinal length varied depending on site variability (see text for details). Every tree (>10 cm DBH) within the overall plot was measured and assigned to a 5 × 5 m plot. These 25 m² plots were clustered together for the 10 × 10 m and 20 × 20 m resolutions. Vegetation plots only occurred at the intersections of every four 5 × 5 m plots.

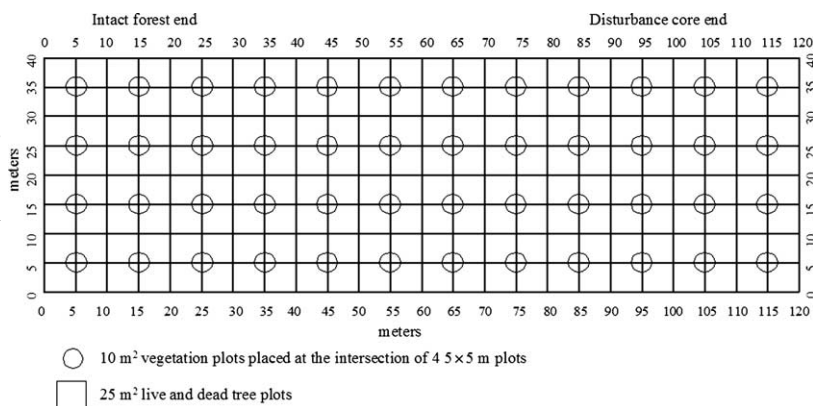


Table 1. Summary and properties of sampled sites.

Site #	Type	Sampling transect (m)		# plots per site at each plot size			Disturbance	Reported disturbance
		Length	Width	5 m	10 m	20 m	year	size (ha)
M02	MPB	110	40	176	44	11	1995	24
M03	MPB	100	40	160	40	10	1993	35
M05	MPB	170	40	272	68	17	1995	133
M06	MPB	120	40	192	48	12	1993	39
M08	MPB	140	40	224	56	14	1995	59
M09	MPB	120	50	240	60	12	1993	171
M10	MPB	130	40	208	52	13	1995	62
M13	MPB	120	40	192	48	12	1993	73
M14	MPB	150	40	240	60	15	1995	75
M15	MPB	140	40	224	56	14	1994	18
M16	MPB	140	40	224	56	14	1994	48
F11	Fire	120	40	192	48	12	1997	62
F12	Fire	160	40	256	64	16	1994	77
F17	Fire	100	60	240	60	15	1994	50
F18	Fire	60	40	96	24	6	1994	6
F19	Fire	100	40	160	40	10	1994	47
F20	Fire	110	40	176	44	11	1994	77
F21	Fire	130	40	208	52	13	1994	50

To characterize the functional response of the vegetation community and the litter layer to the delineated boundary zones, we estimated percent cover of all plant species in 10 m^2 circular plots at the intersection of each group of $4.5 \times 5\text{ m}$ tree plots (the center of each $10 \times 10\text{ m}$ tree plot). We also measured depth of the litter layer to mineral soil.

Boundary delineation

A standardized measure of disturbance severity was used (Basal Area killed/[BA killed + BA alive]) for all boundary delineation and description. This measure was used so that the boundaries would be defined solely by the direct effects of disturbance on trees. To describe the boundaries and the zonation within the transects, we used spatially constrained clustering (hereafter SCC) with the k-means adjustment (Fortin and Drapeau 1995) using the software BoundarySeer (Anon. 2001c). SCC is a technique that creates hard boundaries between dissimilar features creating spatially distinct patches. The hardness or softness of a boundary is scale dependent: what appears hard at one scale may be soft at a finer scale. By providing a complete delineation of a study area, the resulting patches generated during the process represent areas of spatial homogeneity (Fortin and Drapeau 1995) in an area of overall heterogeneity. Given that we were interested in measuring the width and shape of a boundary, the sampling transects needed to be completely delineated. Since we know that a boundary exists at each site because we sampled at the edge of a disturbance, other boundary techniques such as wombling (Fortin and Drapeau 1995) that generate boundary elements would not be appropriate. Boundary detection algorithms are appropriate when the goals of the paper are to find boundaries in a region.

In SCC, formation of clusters using the dissimilarity matrix ($1 - \text{Euclidean distance}$) was constrained to only those plots that are contiguous, creating closed areas with hard boundaries. At each step, the most similar of all spatially adjacent sampling units, or clusters of sampling units, were merged, and coalescing continued until an optimal number of clusters was found using the goodness-of-fit statistic (Gordon 1999). A connectedness of 0.8 was used to minimize single plot clusters (also see Hall and Maruca 2001). For large sample sizes like those that occurred with the 5 m plots (ca 200 plots), there were often several local goodness-of-fit optima. We selected the strongest one where there was a clear, local optimum. An optimum is defined as a goodness-of-fit value that is locally maximized (Fig. 2). Where several were equally good, we used the cluster number that minimized single plot clusters. This usually occurred above 10–15 clusters and below 25. For 10 m plots, there

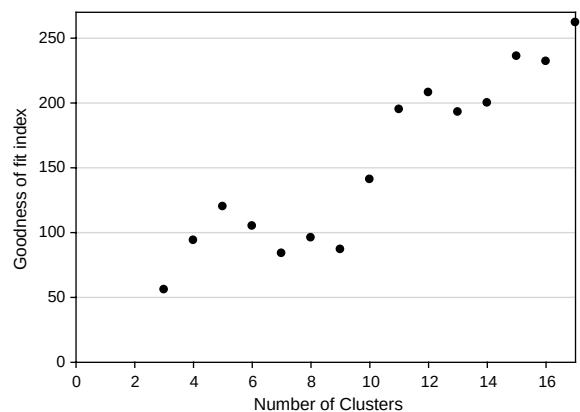


Fig. 2. Sample goodness-of-fit graph. Site M13, 10 m resolution. In this sample, 12 clusters were selected. Alternatively, 5 clusters could have been selected, but mostly single plot clusters would have been created.

was usually only one optimum. For 20 m plots, there was often no optimum; 3 clusters were used as a default for 20 m resolution. Here, a goodness-of-fit optimum at 2 clusters would indicate a hard boundary.

Boundary characterization

Distances between the lowest severity and highest severity clusters were measured using ArcView[®] geographic information system (Anon. 1999; see Fig. 3). These lowest and highest severity clusters were identified at each site to represent areas inside and outside the disturbance, respectively. Clusters that were identified as high or low severity but were $<400\text{ m}^2$ ($<1\%$ of the area at each sites) were rejected as being too small to represent the disturbance core or intact forest. Only larger, contiguous areas were used to represent these extremes of severity. To measure distances between the high and low severity clusters, all sampled plots that were within the cluster and that were adjacent to the cluster edge that was facing the opposite severity cluster were identified (Fig. 3: plots with grey lines through them). These plots create the inner and outer boundaries and are the basis of dividing the area into three zones: inside the disturbance core (DC), in the intact forest (IF) and in the intermediate boundary zone (BZ). The distance was calculated between each plot center along this edge to the nearest plot in the opposite severity cluster. This process was repeated for both low and high severity cluster plots. In cases where there was more than one identified low or high severity cluster (a situation that arose often because of the inherent variability of the boundaries), up to four candidate clusters were selected (two inside the disturbance and two outside) to minimize and maximize the boundary width calculations (Fig. 3A: light grey lines connect plots used to calculate minimum, and dark grey lines connect plots used to calculate maximum). These two calculations are herein referred to as either maximized boundary zone (MaxBZ) or minimized boundary zone (MinBZ). All results use both of these calculations of boundary width. Due to small sample sizes (fires: $n=7$; MPB: $n=11$), comparisons between MPB and fire responses use non-parametric statistics (Anon. 2000).

To measure the heterogeneity in the boundary zones, within-zone variance of standardized disturbance severity was calculated for each of the three zones delineated by the spatially constrained clustering. These levels of variance were compared within and between disturbance types for both MinBZ and MaxBZ. Steepness was simply mean percent change in mortality per linear meter across the boundary zone.

From the vegetation plots, we calculated number of species per unit area – “species richness” (species density, sensu Simpson 1964) – Shannon-Wiener diversity

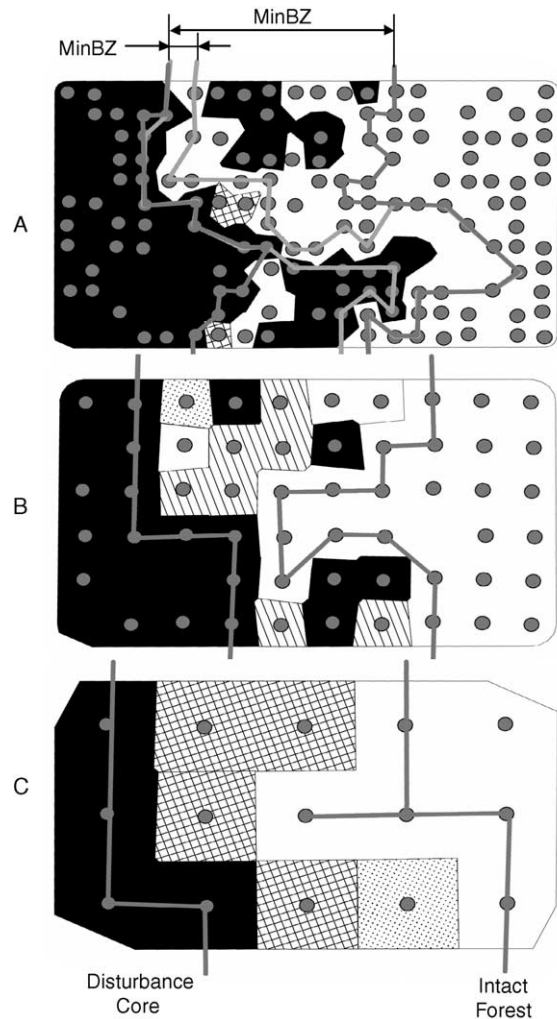


Fig. 3. Cluster maps of one site (F17) showing three resolutions and MaxBZ and MinBZ for each resolution. Cluster design ranges from white (IF), dots, stripes, cross hatching, to black (DC), representing 0–20, 20–40, 40–60, 60–80 and 80–100% mortality. Heavy points are centroids of plots that had trees. Outer boundary (separating IF and BZ) and inner boundary (separating BZ and DC) are represented with grey lines. Light grey lines connect points that were used to calculate MinBZ and dark grey lines connect points that were used to calculate MaxBZ. At this site, only the 5 m resolution had different MaxBZ and the MinBZs. A) 5 m plots; B) 10 m plots; C) 20 m plots.

index and evenness index, total vegetation cover, and the multivariate community structure using the first two axes off a partial detrended canonical correspondence analysis (DCCA; ter Braak 1988, ter Braak and Smlauer 1999). We used a partial analysis to remove the covariance structure related to the spatial proximity of sample plots. These variables were calculated for each of the three zones (IF, BZ, and DC) and compared statistically (generalized linear model single factor unbalanced ANOVA: Proc GENMOD, log-normal distri-

bution for continuous variables and log-Poisson distribution for count variables, Anon. 2000). In addition, we have indicated where the shape of the pattern was biphasic (i.e. increasing then decreasing, or vice versa, as one moves along the transect). Significance for these tests was set to $p < 0.05$, however, the results from the significance tests will be affected in two opposing ways due to the spatial covariance structure across the boundaries. First, because of lack of independence of data points, the effective degrees of freedom is less than the number of sample plots, thereby reducing the power of the statistical tests. In contrast, estimates of mean values in each zone (IF, BZ and DC) will be overly similar to one another because the plots used to calculate the mean values in each zone are in immediate proximity to one another, rather than random samples of the zones. For these contrasting reasons, we discuss the numerical trends in addition to the statistical trends. To examine the first and second axes of the vegetation DCCA simultaneously, we used a multivariate ANOVA to determine whether the plant communities varied across the boundaries (Proc GLM, MANOVA statement, log transformed data, Anon. 2000).

We quantified the vertical structural response of the vegetation community across the three zones using the Shannon-Weiner diversity index and evenness index using 5 vegetation heights (<2 m, 2–10 m, tree sub-canopy, tree canopy, and tree dominants). Examination and analyses of these indices were identical to those for vegetation response.

Results

Peak severity (as measured by BA killed/[BA killed + BA alive]) was significantly higher in the fires than MPB at all plot resolutions (Wilcoxon rank sum test; 5 m: $p = 0.02$; 10 m: $p = 0.006$; 20 m: $p = 0.002$). The magnitude of the differences in peak severity depended on the resolution of the sample unit (Fig. 4). Severity was highest for both fire and MPB at the 5 m resolution (0.97 vs 0.77) and was lowest at the 20 m resolution (0.89 vs 0.43). The three spatial scales were significantly different

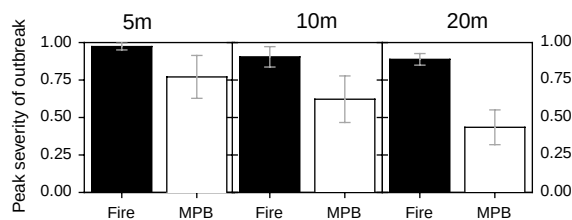


Fig. 4. Peak severity of disturbance event based on the maximum severity clusters (BA killed/[BA killed + BA alive]). Black = fires; white = MPB. Error bars indicate 95% confidence limits (see Table 1 for sample sizes).

for each MPB and fire sites (Kruskal-Wallis tests; MPB: $p = 0.005$, fire: $p = 0.02$).

Then, to determine the number of boundaries to use, the goodness-of-fit analysis indicated that at all resolutions the optima were reached with more than two zones, indicating that there were no purely hard boundaries at these scales (e.g. Fig. 2). This resulted in more complex and convoluted boundary zones (Fig. 3). The three example maps show the boundary zones delineated at a single site at the three resolutions.

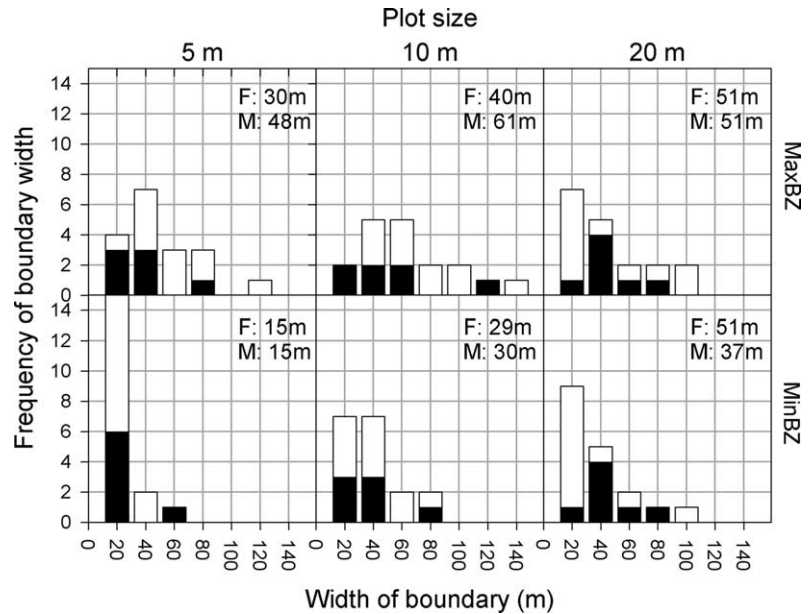
Boundary widths for MPB ranged from 5 m (MinBZ at site M09) to 127 m (MaxBZ at site M05), with MaxBZ means of 48, 61, and 51 m (5, 10, and 20 m plots, respectively, see Fig. 5). Fire boundary widths ranged from 0 m (no boundary zone at site F18) to 107 m (MaxBZ at site F12), with mean MaxBZ widths of 30, 40, and 51 m (5, 10, and 20 m plots, respectively). Mean MinBZ widths for MPB were 15, 30 and 37 m and for fires were 15, 29, and 51 m for the three plot sizes. These large ranges indicate large variability within disturbance types and with distance estimation type. At all resolutions and both MinBZ and MaxBZ estimates, fire and MPB had similar boundary widths with the greatest difference occurring for the MinBZ at the 20 m-plot resolution (Wilcoxon rank sum tests: MPB = 37 m, fire = 51 m, $p = 0.04$). Within each disturbance type, sampling unit resolution did not significantly affect boundary width for MaxBZs (MPB: $p = 0.51$; fire: $p = 0.06$) but was a highly significant influence on boundary width for the MinBZs (MPB: $p = 0.005$; fire: $p = 0.007$), with finer scale sampling producing shorter. (Sample sizes for 20 m resolution are lower than adequate, but we include those data because the results are consistent with the other resolutions and 20×20 m plots are commonly used in plant ecology and forestry applications.)

At all sampling unit resolutions and for both MinBZ and MaxBZ widths, fires had steeper boundaries (Fig. 6). Wilcoxon rank sum tests indicate that MaxBZ steepness at 10 and 20 m-sampling units were significantly steeper for fires than MPB.

In all situations (fires and MPB, at all resolutions), boundary zones showed the highest or equally high variability of tree mortality as compared to zones inside and outside the disturbances (Fig. 7). This pattern was strongest for fires with the boundary zone accounting for the greatest variability at the 5×5 m resolution (for MaxBZ) and at the 10×10 m (for both MinBZ and MaxBZ). MPB showed differences in variance between the three zones at 5 and 10 m (only MaxBZ). MPB showed high variance inside the disturbance and in the boundary zones. In general, variance was lowest at the 20 m resolution and boundary zones were most variable at the 5 m resolution.

Vegetation species richness, diversity, evenness and multivariate community types tended to show a biphasic

Fig. 5. Histogram of boundary widths by plot size, disturbance type, and estimation type. Means are indicated on each graph (M: MPB, F: Fire). Black = fire; white = MPB. Wilcoxon rank sum tests, *: $p < 0.05$.



pattern across most boundaries for both fires and MPB outbreaks at both descriptions of the boundary zone (MinBZ and MaxBZ, Table 2). Some of these trends were not statically significant (at $p < 0.05$). Boundary zone vegetation cover tended to be intermediate between the IF and DC (those which are not biphasic). Likewise, foliage height diversity and evenness in the boundary zone tended to be intermediate between the IF and DC at most sites (Table 2). On average, over 50% of variables measured across fire boundaries were significant, whereas ca 1/3 of the variables measured across MPB boundaries were significant. In general, vegetation diversity had a greater tendency to show a biphasic trend across the boundaries and foliage height diversity showed more of a monotonic trend across the boundaries. MinBZ and MaxBZ values for all measured traits differed slightly, but with no apparent tendency.

Multivariate ANOVAs of the first and second DCCA axes of understory vegetation abundances by species showed that approximately half of the sites (8 for MaxBZ and 7 for MinBZ) had a significant change (Wilk's Lambda at $p < 0.05$) in vegetation community across the three zones of the transects.

Discussion

There are three primary results of ecological and conservation significance from these data: 1) mean boundary widths ranged from 0 to 127 m with a mean of 51 m for both MPB and fire when calculated using a 20 m sampling unit resolution; 2) the boundary zones are at least, if not more, heterogeneous in terms of tree mortality than the disturbance area; and 3) across a range of vegetation traits most sites showed a non-linear trend across the boundaries suggesting the ecological importance of boundary zones in forest ecosystem dynamics. This study also demonstrated an objective methodology to delineate zones across forest boundaries that are not traditional "edges". The boundaries in this study ranged in diffuseness of tree mortality (edge effect is usually at abrupt edges, e.g. Murcia 1995, Harper and Macdonald 2001) and are not traditional "ecotones" which are often defined as transition zones between different ecological zones (e.g. Holland et al. 1991; note however, ecotone, as defined by Gosz (1993), provides the correct general context). This study has offered ways to detect the existence of such boundaries which are likely ubiquitous in natural forest ecosystems and to compare these zones between different disturbance types. These methods allowed us to determine how the

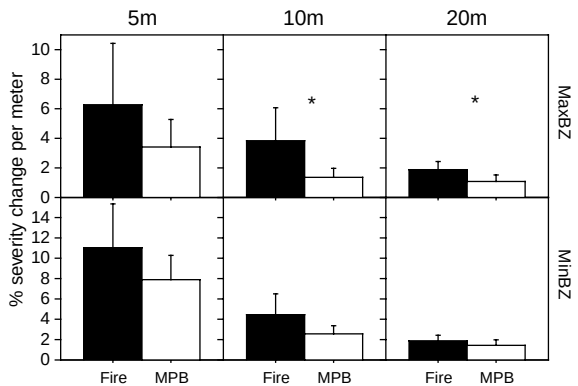


Fig. 6. Steepness of boundaries. Black = fires; white = MPB (units are % severity change per meter $\pm 95\%$ C.I.). Wilcoxon rank sum tests, *: $p < 0.05$.

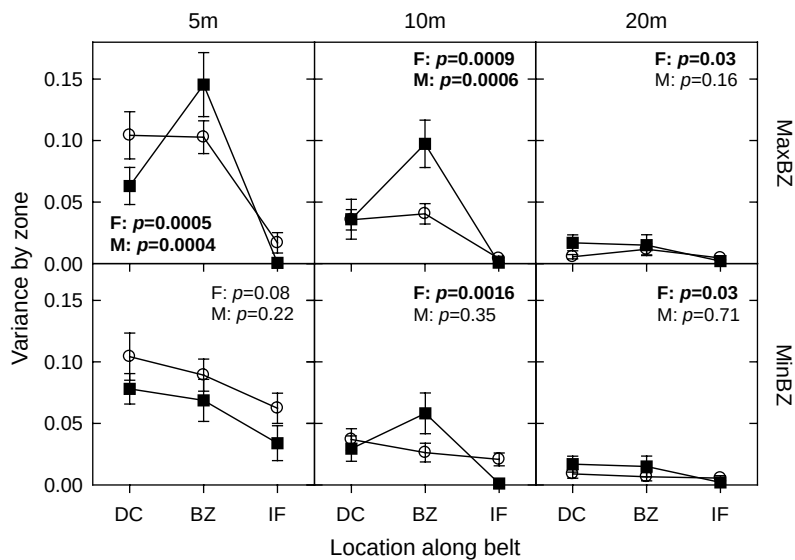


Fig. 7. Between-plot variance in severity of disturbance within each of three zones: inside the disturbance, in the boundary zone (B.Z.) and outside the disturbance. See text for MaxBZ and MinBZ. Probability levels are shown comparing the three zones for fire and MPB (Kruskal-Wallis, Anon. 2000).

magnitude of these boundaries varies in a forest landscape.

The range in boundary width was assessed by comparing the minimum and maximum boundary width values (MinBZ and MaxBZ respectively). While neither width calculation is more correct than the other, they provide different insights about forest boundary zones and vegetation responses to them. In this study we were more interested in determining the size of a region of high heterogeneity, thus, we emphasize the results obtained from the maximum estimated widths (MaxBZ).

Here, our sampling design allowed for direct comparison of spatial patterns at the boundaries of these disturbance types. Indeed, most natural disturbance studies examine only one type of disturbance (Bergeron

et al. 1998, Cissel et al. 1999) and rarely have there been direct comparisons between disturbance types in a single sampling study (though see Foster et al. 1998). Our findings indicated that while boundary demarcation was generally more pronounced in fires than in MPB, both types of natural disturbance produced a range from hard to soft boundaries. Although the two disturbances, fires and MPB, are governed by different processes, the resulting boundary widths were comparable. Other recent work has demonstrated that both fires and MPB boundaries are affected by the amount and presence of susceptible trees (McIntire 2004). These results also showed that peak severity (measured as tree mortality) of the fires was higher than MPB at all sampling unit resolutions (though with a smaller

Table 2. Percentage of sites where vegetation, tree foliage and leaf litter response patterns across the boundary transects was significant (ANOVA, $p < 0.05$) or having a biphasic pattern. Biphasic indicates that the BZ was higher or lower than both disturbance and intact forest (with no test of statistical significance). Min = MinBZ, Max = MaxBZ.

Trait	Percentage of sites with a significant pattern across boundary				Percentage of sites where pattern is biphasic across boundary			
	Fire		MPB		Fire		MPB	
	MaxBZ n = 5	MinBZ n = 5	MaxBZ n = 11	MinBZ n = 10	MaxBZ n = 5	MinBZ n = 5	MaxBZ n = 11	MinBZ n = 10
Veg. sp. richness	40	60	45	40	60	20	55	80
Veg. diversity ^a	80	80	36	10	60	40	55	90
Veg. evenness ^b	40	40	18	10	60	40	45	50
Veg. DCCA axis 1 ^c	40	40	27	40	60	60	91	90
Veg. DCCA axis 2 ^c	0	0	0	10	100	100	73	100
Veg. cover	60	60	9	10	60	20	36	30
Foliage diversity ^d	80	80	36	60	0	60	0	20
Foliage evenness ^e	100	80	36	50	40	40	36	30
LFH depth	40	40	73	70	40	40	36	40
Average	53	53	31	33	53	47	47	59

^aShannon-Weiner diversity index; ^bShannon-Weiner evenness index; ^cPartial detrended canonical correspondence analysis (ter Braak 1988); ^dShannon-Weiner foliage height diversity; ^eShannon-Weiner foliage height evenness.

difference at 5 m resolution). However, the sites in this study likely do not include lowest severity fires (of which there are few in the study area) and outbreaks (as they are difficult to detect from aerial photographs and in the field).

Heterogeneity, as measured by the variance of plots in each zone, varied with quadrat size. As expected, the MaxBZs showed a generally higher level of variability in the boundary zones than the MinBZs. This confirms that the difference between these two measures of boundary width is simply a reallocation of variability into or out of the boundary zone. In the MaxBZs, MPB showed high variance in the boundary zone and inside the disturbance, whereas fires appeared to have most of the variability in the boundary zone. This pattern of high variability in the boundaries is not unexpected, and many people have suggested that natural boundaries are more variable than human induced boundaries (Krummel et al. 1987, Forman 1995, Foster et al. 1998). Methodology quantifying this variability is, however, an important step in the analysis and understanding the spatial dynamics of natural boundaries.

The biphasic vegetation pattern that we found across many of the boundaries for many vegetation features demonstrates that as mortality decreases monotonically across the edge of a disturbance, vegetation community structure does not respond in a parallel manner; intermediate tree mortality is the area of greatest or least vegetation cover for different species. This partial result (i.e. a common trend, though not all significant) suggests that the multivariate vegetation community may not have had enough time since the disturbance to diversify, at some of the sites. When these results on heterogeneity and vegetation response are examined in combination with the tree natural regeneration that demonstrates non-linear successional trajectories in these boundary zones (McIntire unpubl.), it appears that the response of the forest in boundary zones changes landscape functioning as compared to boundary lines (sharp boundaries).

Spatially constrained agglomerative clustering was determined to be an efficient, intuitive and exploratory methodology to describe natural disturbance boundaries. Many other approaches, such as fuzzy set theory, probability mapping, adapted neural networks and fractals (reviewed by Kent et al. 1997), were not suited to field collected data because of the large sample sizes needed. With field data, edge detector algorithms such as wombling (Fortin 1994) and moving split-window (Johnston et al. 1992) have been suggested techniques, though these are optimally used for detection of unknown boundaries (Jacquez et al. 2000). Other analysis tools, such as autocorrelation or semi-variograms, would be able to indicate the direction of the greatest anisotropy and therefore the direction of the boundary, but would not be able to quantify the size of

the boundary. In situations where the presence of a boundary is not in doubt, such as those that occur in natural disturbances, but the nature and description of that boundary is unknown, spatially constrained clustering appears to be efficient and intuitive. As a final note on our choice of methodology, while we determined boundaries using spatially constrained clustering, both boundary width indexes can be calculated using boundaries delineated by other boundary detection methods (Fagan et al. 2003).

Boundary zones of the width measured in this study (ca 50 m) would comprise ca 22% of the area of the disturbance in a hypothetical circular 85 ha disturbed area. However, since no disturbance will be a perfect circle, the perimeter of a natural disturbance will be longer, creating an even larger boundary zone. In a range of disturbances from 40 to 200 ha, the proportion of the area that would be in a boundary zone would be between 14 and 30%. (We remind the reader, however, that we only measured the flanks of fires, and so we do not know what random edges of fires would look like. Since fires can propagate for long distances, flanks may rapidly become the largest part of the perimeter of a fire, so this is not of great concern.) This demonstrates the potential areal importance of these boundary zones and associated conservation implications of such a large component of a landscape in a heterogeneous state that has generally been overlooked. Here, in contrast to a traditionally defined edge effect, we have measured a diffuseness in disturbance severity, not simply a hard edge influence into an intact forest or into a forest opening.

Our findings show that natural disturbance boundaries were highly heterogeneous that in turn resulted in different functional vegetation responses. Boundary structural properties ranged however from hard to soft at all resolutions from 5 to 20 m. While this study does not directly compare natural disturbance boundaries with forest management-induced boundaries, the preponderance of studies examining edge effects of harvest edges suggest that they are abrupt by making the implicit assumption that the edge can be assigned a zero distance value (Chen et al. 1992, Murcia 1995, Fernandez et al. 2002), something that was impossible at most of the boundaries in this study. Furthermore, few management plans of which we are aware create a range of boundary contrasts as those found in this study. In many managed forest ecosystems where forest management polygons range from < 40 to 200 ha (though sometimes greater), boundary zones would comprise from 14 to over 30% of a landscape area. Implementation of spatial structure of boundaries in managed landscapes, following the natural disturbance coarse filter strategy (e.g. Swetnam et al. 1999), may be important to maintain ecological processes of forested ecosystems both at the stand (e.g. facilitate forest regeneration) and landscape (e.g. favoring biodiversity) levels.

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